

What are ecosystem services?

Ecosystems are complex interactions among living and non living components of the environment (e.g., forests, grasslands, marine ecosystems). These interactions mediate processes that achieve major transformations of resources, many rivalling or exceeding what can be cost effectively achieved by humans (e.g. regulation of atmospheric gases, large scale filtration and purification of water). These transformations (ecosystem services) support and enrich human life, but are often overlooked in decision making because decision makers lack information about them, and they (mostly) are outside economic markets and haven't had an economic value attached to them.

Ecosystem services have been defined as the benefits to humans from nature or, direct and indirect contributions of ecosystems to human wellbeing. For the purposes of '[environmental accounting](#),' ecosystem services have been distinguished further as those that can be turned directly into benefits (called 'final ecosystem services') and those that support other services (called 'intermediate ecosystem service'). While still far from perfect, these distinctions are increasingly gaining acceptance in assessment processes around the world.

The ecosystem services approach

The 'ecosystem services approach' is an approach that considers the full range of benefits and costs (both monetary and non monetary) associated with decisions that affect, or are affected by, ecosystems. That is, ecosystems provide outputs and outcomes that directly and indirectly affect human wellbeing, and these considerations link to a policy approach that seeks to make informed trade-offs between social, environment and economic elements. The ecosystem services approach is seen as providing a solid basis for discussing, across government and society, the challenges and opportunities for human interactions with the environment. The ecosystem services approach to natural resources management is now being used by governments in the United Kingdom, the United States, Canada and New Zealand.

The Department of Agriculture Fisheries and Forestry (DAFF) commissioned [Australia 21](#) to produce a report on how ecosystem services concepts are evolving, and their relevance to policy development for the Australian Government and other sectors of society, with a particular focus on their application to agricultural lands. The report, "[Discussion paper on Ecosystem Services for the Department of Agriculture, Fisheries and Forestry](#)", concluded that one of the greatest benefits of the adoption of an ecosystem service approach would be its value in facilitating dialogue between parties on complex major policy challenges facing Australia. It recommends building on existing approaches and developing tools and improving governance arrangements to facilitate assessment of ecosystem services across all areas of government and society.

Ecosystem services and agricultural land

Ecosystems include agricultural landscapes which are particularly important in the Australian context; they occupy more than 60 per cent of the land, and actions taken by farmers on these landscapes have a

significant impact on the quality of ecosystem services delivered to the community. For example, the way dairy farmers dispose of milking shed waste and their fertiliser practices have a big impact on the water quality; nutrients lost to streams and lakes increase algal blooms which cost the community through: increased need for water treatment, adverse impacts on stream biota (fish kills), amenity, recreation and tourism opportunities (and associated economic losses) because the sight and smell is unpleasant and contact with the algae is dangerous to human and animal health.

DAFF, through Caring for our Country, has approved \$448 million (up to 30 May 2012) for projects to improve land management practices on farm [to enhance soil condition](#) and as a consequence the quality of ecosystem services delivered from agricultural lands to the broader community. The report, [“The relationships between land management practices and soil condition \(wind and water erosion, soil pH and carbon\) and the quality of ecosystem services delivered from agricultural land in Australia”](#) shows that there is good evidence from scientific research that the land management practices adopted by farmers have a direct impact on soil condition. Improving soil condition benefits production as well as providing a range of ecosystem services to the broader community; including water purification, breakdown of wastes and toxins, regulation of atmospheric gases (including carbon and nitrogen) and water flows, regulation of weather and climate and maintenance of genetic diversity.

Further reading

Maynard S., James D. & Davidson A. (2010) The Development of an Ecosystem Services Framework for South East Queensland. *Environmental Management* 45, 881-95.

Millennium ecosystem assessment 2005

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/natural-resources/ecosystem-services>